Controller Design for Flapping Flight

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We propose a control strategy for flapping flight.

2 Model Description

The flapping flight dynamics model consists of a point mass body, free to move in two dimensions, subjected to gravity and to the aerodynamic force from N wing pairs. The two spatial directions are $\hat{x}$, which is parallel to the ground, and $\hat{z}$, which is parallel and opposed to the direction of gravity. The spatial coordinates (x^*, z^*) are nondimensionalized by the body length L. Time t^* is nondimensionalized by the gravitational time scale $\sqrt{L/g}$. Each wing pair is treated as massless, and has a dimensionless inverse wing loading $(A^*/m^*)/N$, where

$$\frac{A^*}{m^*} = \frac{\rho_{\text{air}} L A}{m} \tag{1}$$

in which m is the body mass and A the total wing surface area. The dimensionless value of the weight mg is 1. In our 2D approximation, both wings of each pair are lumped into a single surface element, free to translate along the stroke axis $\hat{s}$. The translational coordinate is s^*. The stroke basis $(\hat{s}, \hat{n})$, consisting of the stroke axis $\hat{s}$ and the stroke-plane normal vector $\hat{n}$, is obtained by rotating the fixed frame basis $(\hat{x}, \hat{z})$ by the stroke plane angle γ. The angular sign convention in the model is that a positive angle results in a counterclockwise rotation. The flapping motion of each wing pair element is prescribed by the kinematic law

$$s^*(t^*) = s_0^* \sin \omega^* t^* \implies s^*(\tau^*) = s_0^* \sin \tau^* \tag{2}$$

where s_0^* and ω^* are the (dimensionless) amplitude and frequency parameters, respectively. We will use the shorthand $\tau^* = \omega^* t^*$, denoting the phase. A wingbeat cycle occurs between $\tau^* = 0$ and $\tau^* = 2\pi$.

Each wing pair element is oriented perpendicular to the $(\hat{x}, \hat{z})$ plane, and is free to rotate or *pitch* about its aerodynamic center. The pitch angle ψ is defined relative to the stroke-plane normal $\hat{n}$. The pitch kinematics are prescribed by

$$\psi(\tau^*) = \psi_0 + \psi_1 \sin(\tau^* - \delta_0) \tag{3}$$

where ψ_0 is the mean pitch angle, ψ_1 is the pitch amplitude, and δ_0 is the phase offset relative to the flapping motion. The quantities relevant to the wing kinematics, namely γ, s^*, and ψ, are represented for a single wing pair element in Figure 1. From these quantities and the body velocity vector, the angle of attack α can be determined, along with the lift and drag force directions. This is discussed in more detail in the following section. To finally obtain the aerodynamic force on each wing pair element, a lift and drag coefficient model is used, following [1, 2], which gives

$$C_L = C_{L,0} \sin 2\alpha \tag{4}$$

$$C_D = C_{D,0} \cos^2 \alpha + C_{D,\frac{\pi}{2}} \sin^2 \alpha \tag{5}$$

where $C_{L,0} = 1.5$, $C_{D,0} = 0.1$, and $C_{D,\frac{\pi}{2}} = 2.0$. An important simplification of the model is that the aerodynamic moments on the body are neglected, and there are no attitude dynamics.

When $N = 2$, to model a dragonfly for example, there is an additional phase parameter σ_0 representing the phase offset between the two wing pairs, which may then beat in sync, out of phase, or somewhere in between. The kinematics of the two wing pairs remain otherwise identical.

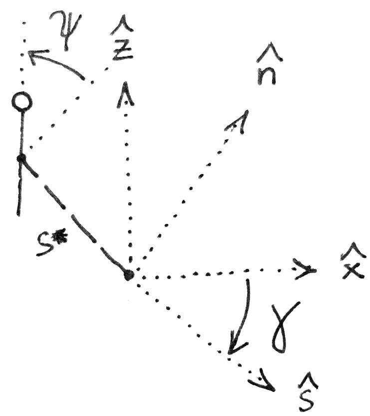

Figure 1: Definition of the wing kinematic quantities γ, s^*, and ψ. As represented, $\gamma < 0$, $s^* < 0$, and $\psi > 0$.

3 Controller Design

3.1 Problem Statement

We wish to design a control strategy to achieve arbitrary trajectories in the two-dimensional plane, including prey pursuit trajectories. We assume that the flyer's own state is known using a variety of sensory inputs, so position and velocity are available as inputs to the controller. In addition, when a prey is present in the environment, the bearing angle to the prey, as detected by the eyes, is also an available input. Because we are particularly interested in *dragonfly* flight, we make two design decisions such that the resulting flight characteristics resemble those of a dragonfly. First, hovering should be achieved with an inclined stroke plane (tilted from the horizontal), characterized by the parameter γ^{hover}. Second, when at speed, the stroke-plane normal should be approximately aligned with the direction of travel, as observed in experiments [3]. This last choice will simplify the analysis to follow, since any flow velocity component parallel to the stroke plane can be neglected. With these choices in mind, the contents of this section otherwise hold for flapping flight in general, as we do not assume specific morphological characteristics like number of wing pairs, size, etc.

3.2 Mean Aerodynamic Force

3.2.1 Definitions and Assumptions

To affect its trajectory, the flying body must modulate and direct the aerodynamic force acting on its wings. Because the aerodynamic force on a flapping wing is by nature highly unsteady, it is convenient to consider the wingbeat-cycle average of the aerodynamic force, summed over each of the wings, which we call the *mean aerodynamic force*, $\langle \vec{F}^* \rangle$. $\langle \vec{F}^* \rangle$ is a function of morphology and wing kinematics, as well as the flow characteristics in the body frame (this involves the body kinematics in the fixed frame, as well as any wind in the fixed frame; in the following we refer to this simply as the body kinematics, or body velocity, with the implication that it is relative to the air, which need not be the same as the fixed frame if there is any wind). Body velocity is not constant over a wingbeat cycle: there are (a) oscillations inherent to flapping, and there may be (b) a net change in body velocity over the cycle. Assuming the oscillations are small compared to the characteristic wing velocity, and that any body velocity changes occur at a slow rate relative to the wingbeat period, the body velocity may be considered constant over a wingbeat cycle. The following assumes these conditions are met. Our objective is now to understand how $\langle \vec{F}^* \rangle$ depends on the wing kinematics at any given body velocity, so that we are able to formulate a control strategy.

As explained in subsection 3.1, we are designing a controller under the assumption that the body velocity is normal to the stroke plane. Beyond morphology and kinematics, then, there is a single body velocity parameter $\langle u^* \rangle$, which is the cycle-averaged body velocity magnitude (the direction being set to the stroke-plane normal $\hat{n}$). It is convenient to introduce a similarity parameter in the form of the *advance ratio*, J, which is the ratio of the cycle-averaged body velocity to the characteristic wing velocity:

$$J = \frac{\langle u^* \rangle}{s_0^* \omega^*} \tag{6}$$

The instantaneous wing velocity $\vec{v}^*$ relative to the air has the stroke-plane normal and in-plane components

$$v_n^* = \langle u^* \rangle = s_0^* \omega^* J \tag{7}$$

$$v_s^* = s_0^* \omega^* \cos \tau^* \tag{8}$$

The squared norm of the velocity is

$$(v^*)^2 = \left(s_0^* \omega^*\right)^2 \left[J^2 + \cos^2 \tau^*\right] \tag{9}$$

We introduce the signed angle χ from $\hat{n}$ to $\vec{v}^*$, defined as

$$\chi = -\tan^{-1} \frac{v_s^*}{v_n^*} = -\tan^{-1} \frac{\cos \tau^*}{J} \tag{10}$$

When $\cos \tau^* > 0$ (stroke along $+\hat{s}$, $\chi < 0$), the angle of attack α is

$$\alpha = \psi - \chi = \psi + \tan^{-1} \frac{\cos \tau^*}{J} \tag{11}$$

When $\cos \tau^* < 0$ (stroke along $-\hat{s}$, $\chi > 0$), the angle of attack α is

$$\alpha = \chi - \psi = -\psi - \tan^{-1} \frac{\cos \tau^*}{J} \tag{12}$$

The general expression is, then,

$$\alpha = \frac{\cos \tau^*}{|\cos \tau^*|} \psi + \tan^{-1} \frac{|\cos \tau^*|}{J} \tag{13}$$

The above is sufficient to obtain the mean aerodynamic force. The derivation is somewhat involved in the general case. We first present the hovering case, which corresponds to $J = 0$.

3.2.2 Hover Case ($J = 0$)

For $J = 0$, the angle of attack simplifies to

$$\alpha = \frac{\cos \tau^*}{|\cos \tau^*|} \psi + \frac{\pi}{2} \tag{14}$$

Using the appropriate trigonometric identities, the aerodynamic coefficients (defined in Equation 4 and Equation 5) are

$$C_L = -\frac{\cos \tau^*}{|\cos \tau^*|} C_{L,0} \sin 2\psi \tag{15}$$

$$C_D = \frac{C_{D,\frac{\pi}{2}} + C_{D,0}}{2} + \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \cos 2\psi \tag{16}$$

Because the wing velocity relative to the air is purely in-plane (along $\hat{s}$), the in-plane aerodynamic force component is purely drag, and the out-of-plane component (along $\hat{n}$) is purely lift. The mean aerodynamic force components are

$$\begin{aligned}
\langle F_n^* \rangle &= \frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \langle C_L |\cos\tau^*|^2 \rangle \\
&= -\frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 C_{L,0}\langle \sin 2\psi \;\; \cos\tau^* |\cos\tau^*| \rangle
\end{aligned} \tag{17}$$

$$\begin{aligned}
\langle F_s^* \rangle &= -\frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \langle C_D \cos\tau^* |\cos\tau^*| \rangle \\
&= -\frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \langle \cos 2\psi \;\; \cos\tau^* |\cos\tau^*| \rangle
\end{aligned} \tag{18}$$

Note the constant drag term vanishes because $\langle \cos\tau^* |\cos\tau^*| \rangle = 0$. We introduce the shorthand $\Delta C_D = C_{D,\frac{\pi}{2}} - C_{D,0}$. The expressions feature a common pre-factor we call q^*, defined as

$$q^* = \frac{1}{4}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2 \tag{19}$$

We further introduce the normal and in-plane force coefficients $C_{F_n^*}$ and $C_{F_s^*}$, defined as

$$C_{F_n^*} = \frac{F_n^*}{q^*} = -2C_{L,0}\langle \sin 2\psi \;\; \cos\tau^* |\cos\tau^*| \rangle \tag{20}$$

$$C_{F_s^*} = \frac{F_s^*}{q^*} = -\Delta C_D \langle \cos 2\psi \;\; \cos\tau^* |\cos\tau^*| \rangle \tag{21}$$

which we synthesize into the overall force magnitude coefficient

$$C_{F^*} = \sqrt{C_{F_n^*}^2 + C_{F_s^*}^2} \tag{22}$$

This coefficient is useful because it is only a function of the pitch kinematic parameters and the lift and drag coefficient parameters, and is independent of morphology and wingbeat frequency and amplitude, which appear only in the pre-factor q^*. By maximizing the value of this coefficient, given a constraint on the force direction required to achieve a given maneuver, the characteristic wingbeat velocity $s_0^*\omega^*$ can be minimized. With our parametrization of the pitch kinematics, and with the lift and drag coefficient parameters set, C_{F^*} and its (n, s) decomposition are functions of 3 variables: ψ_0, ψ_1, and δ_0. We visualize C_{F^*}, $C_{F_n^*}$, and $C_{F_s^*}$ across slices of this 3D space in Figure 2 and Figure 3.

As panel (a) of Figure 3 makes evident, significantly more force is produced when $\delta_0 = \pm 90°$. Because $\delta_0 = -90°$ results in the mean force pointing along $-\hat{n}$ when the mean pitch is neutral ($\psi_0 = 0°$), we select $\delta_0 = +90°$ as the operating point. This choice maximizes the mean force for a given (ψ_0, ψ_1), and reduces the pitch parameter space to 2D. How then do we select ψ_0 and ψ_1? It is clear from panel (b) that a certain value of ψ_1, around $\psi_1 \approx 50°$, maximizes C_{F^*} for any given ψ_0. We will call it ψ_1^{opt}. In addition, $\psi_0 = 0°$ (mod $90°$) maximizes C_{F^*}. Looking at panel (b) of Figure 2, it is clear that the in-plane force component is zero when $\psi_0 = 0° \pmod{90°}$, so the force is purely normal to the stroke plane. In addition, from panel (a) of the same figure, we see that $\psi_0 = 0°$ results in the mean force pointing towards $+\hat{n}$, while $\psi_0 = \pm 90°$ results in the mean force pointing towards $-\hat{n}$. We select neutral pitch ($\psi_0 = 0°$) as the reference, for consistency with the choice of $\delta_0 = 90°$. In summary, the reference pitch kinematics

$$\psi_0 = 0° \qquad \psi_1 = \psi_1^{\text{opt}} \qquad \delta_0 = 90° \tag{23}$$

result in $C_{F^*} = C_{F^*}^{\max}$, and $\langle \vec{F}^* \rangle$ pointing towards $+\hat{n}$. These reference kinematics allow for *efficient* hover (in the sense that the characteristic wingbeat velocity $s_0^*\omega^*$ is minimized, because C_{F^*} is maximized) when the

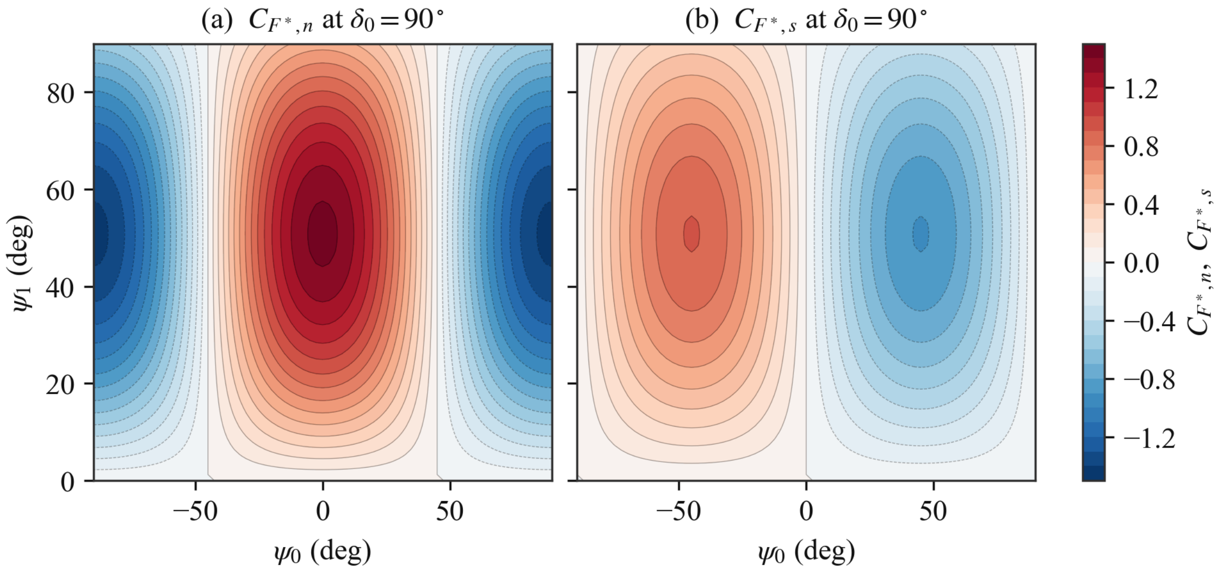

Figure 2: (a) Stroke-plane-normal (lift) component $C_{F_n^*}$ and (b) in-plane (drag) component $C_{F_s^*}$ of the force coefficient over the pitch plane at $\delta_0 = 90°$. The force direction quadrant in the $(\hat{s}, \hat{n})$ plane can be inferred from the signs.

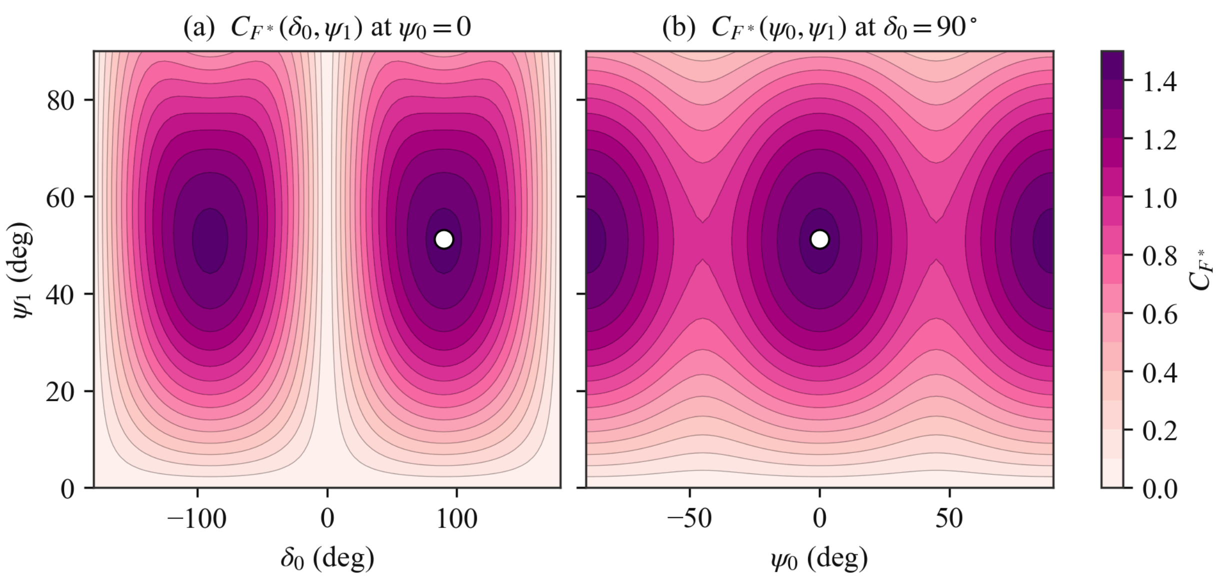

Figure 3: Dependence of C_{F^*} on pitch parameters at zero body velocity. Maximum shown as white dots.

stroke plane is horizontal ($\gamma^{\text{hover}} = 0°$). Indeed, since the mean force vector points towards $+\hat{n}$, $+\hat{n}$ must be vertical and opposed to the direction of gravity. To model dragonfly hover specifically, at an inclined stroke plane ($\gamma^{\text{hover}} \neq 0°$), the mean force vector must have an in-plane component. Looking at panel (b) of Figure 2, it is clear that this can be achieved by changing ψ_0, away from the symmetric 0° point. For convenience, we introduce β, the signed angle from the stroke-plane normal $\hat{n}$ to the mean force vector. Since the mean force is $\langle \vec{F}^* \rangle = q^* \left(C_{F_n^*} \hat{n} + C_{F_s^*} \hat{s} \right)$, β can be expressed from the $C_{F_n^*}$ and $C_{F_s^*}$ coefficients (Equation 20 and Equation 21) as

$$\beta = -\tan^{-1} \frac{C_{F_s^*}}{C_{F_n^*}} = -\tan^{-1} \left[\frac{\Delta C_D}{2C_{L,0}} \frac{\langle \cos 2\psi \cos \tau^* |\cos \tau^*| \rangle}{\langle \sin 2\psi \cos \tau^* |\cos \tau^*| \rangle} \right] \tag{24}$$

We can simplify the $\langle \cos 2\psi \cos \tau^* |\cos \tau^*| \rangle / \langle \sin 2\psi \cos \tau^* |\cos \tau^*| \rangle$ ratio in the above expression. With $\delta_0 = 90°$, the pitch kinematics reduce to $\psi = \psi_0 - \psi_1 \cos \tau^*$. Using the appropriate trigonometric identities,

$$\cos 2\psi = \cos 2\psi_0 \cos \left(2\psi_1 \cos \tau^* \right) + \sin 2\psi_0 \sin \left(2\psi_1 \cos \tau^* \right) \tag{25}$$

$$\sin 2\psi = \sin 2\psi_0 \cos \left(2\psi_1 \cos \tau^* \right) - \cos 2\psi_0 \sin \left(2\psi_1 \cos \tau^* \right) \tag{26}$$

Under the half-period shift $\tau^* \to \tau^* + \pi$, the factors $\cos \tau^* |\cos \tau^*|$ and $\sin \left(2\psi_1 \cos \tau^* \right)$ change sign while $\cos \left(2\psi_1 \cos \tau^* \right)$ is unchanged, so the $\cos \left(2\psi_1 \cos \tau^* \right)$ terms average to zero, leading to

$$\langle \cos 2\psi \;\; \cos \tau^* |\cos \tau^*| \rangle = \sin 2\psi_0 \langle \sin \left(2\psi_1 \cos \tau^* \right) \cos \tau^* |\cos \tau^*| \rangle \tag{27}$$

$$\langle \sin 2\psi \;\; \cos \tau^* |\cos \tau^*| \rangle = -\cos 2\psi_0 \langle \sin \left(2\psi_1 \cos \tau^* \right) \cos \tau^* |\cos \tau^*| \rangle \tag{28}$$

The common $\langle \sin \left(2\psi_1 \cos \tau^* \right) \cos \tau^* |\cos \tau^*| \rangle$ factor cancels out in the ratio, giving simply

$$\frac{\langle \cos 2\psi \;\; \cos \tau^* |\cos \tau^*| \rangle}{\langle \sin 2\psi \;\; \cos \tau^* |\cos \tau^*| \rangle} = -\tan 2\psi_0 \tag{29}$$

and Equation 24 becomes

$$\beta = \tan^{-1} \left[\frac{\Delta C_D}{2C_{L,0}} \tan 2\psi_0 \right] \tag{30}$$

Importantly, β is independent of ψ_1, and is only controlled by ψ_0. It follows that ψ_0 is an appropriate control variable for the mean force direction relative to the stroke-plane normal $\hat{n}$. β is shown as a function of ψ_0 in Figure 4. As discussed above, C_{F^*} becomes less than its maximum when $\psi_0 \neq 0° \pmod{90°}$, away from the reference pitch kinematics, so it is also shown in addition to β. In the $\Delta C_D / \left(2C_{L,0} \right) \to 1$ limit, the relation between β and ψ_0 can be approximated as $\beta \approx 2\psi_0$. This holds only roughly here because $\Delta C_D / \left(2C_{L,0} \right) = 1.9/3 \approx 0.63$, but the trend is the same, as can be seen in the figure. To achieve hover at an inclined stroke plane with tilt angle γ^{hover}, the mean force vector angle must be $\beta = -\gamma^{\text{hover}}$. This requires $\psi_0 = \psi_0^{\text{hover}}$, given by Equation 30 (specifically the inverse relation) when $\beta = -\gamma^{\text{hover}}$.

Now that we have narrowed down the pitch kinematics for hover, we step back to consider the flapping kinematics as well, governed by parameters s_0^* (amplitude) and ω^* (frequency). The condition for hover on the mean force magnitude is

$$C_{F^*} q^* = 1 \implies \left(s_0^* \omega^* \right)^2 = \frac{4}{C_{F^*} \frac{A^*}{m^*}} \tag{31}$$

We prefer to consider ω^* as a fixed parameter rather than a control variable, leaving s_0^* as the primary control for the force amplitude. The hover value is given by

$$\left(s_0^* \right)^{\text{hover}} = \frac{1}{\omega^*} \sqrt{\frac{4}{C_{F^*}^{\text{hover}} \frac{A^*}{m^*}}} \tag{32}$$

The hover kinematic parameters are summarized in Table 1. The control variables are s_0^* and ψ_0.

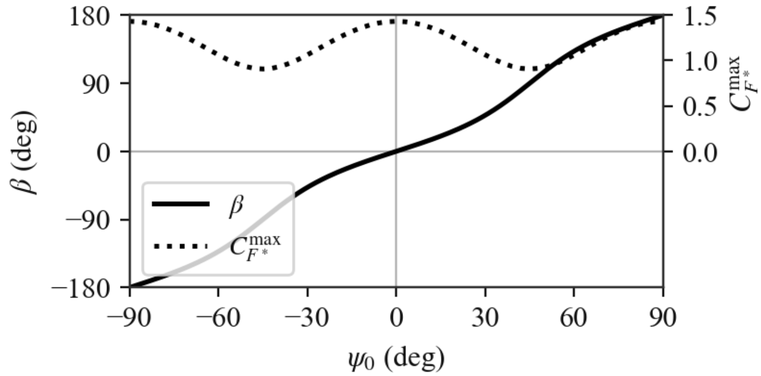

Figure 4: Effect of ψ_0 on force direction angle β and force coefficient C_{F^*}.

Table 1: Wing kinematic parameters for hover

Parameter	Value	Variables				
s_0^*	$(s_0^*)^{\text{hover}}$	$\frac{A^*}{m^*}$	ω^*	γ^{hover}	$C_{L,0}$	ΔC_D
ψ_0	ψ_0^{hover}			γ^{hover}	$C_{L,0}$	ΔC_D
ψ_1	ψ_1^{opt}					None
δ_0	90°					None

3.2.3 General Case ($J \geq 0$)

For $J \neq 0$, the wing velocity relative to the air is no longer purely in-plane, so lift and drag each contribute to both force components. In terms of the angle χ, the wing velocity direction is $\hat{v} = \cos\chi\,\hat{n} - \sin\chi\,\hat{s}$. Drag acts along $-\hat{v}$, and lift acts along the perpendicular direction

$$\hat{\ell} = -\sigma\,(\sin\chi\,\hat{n} + \cos\chi\,\hat{s}) \qquad \sigma \equiv \frac{\cos\tau^*}{|\cos\tau^*|} \tag{33}$$

whose sense alternates between half-strokes, consistent with the sign convention used to define α (in the hover case, $\hat{\ell}$ reduces to $+\hat{n}$ on both half-strokes). The instantaneous force is

$$\vec{F}^* = \frac{1}{2}\frac{A^*}{m^*}(v^*)^2\left(C_L\,\hat{\ell} - C_D\,\hat{v}\right) \tag{34}$$

with components

$$F_n^* = \frac{1}{2}\frac{A^*}{m^*}(v^*)^2\,(-\sigma\sin\chi\,C_L - \cos\chi\,C_D) \tag{35}$$

$$F_s^* = -\frac{1}{2}\frac{A^*}{m^*}(v^*)^2\,(\sigma\cos\chi\,C_L - \sin\chi\,C_D) \tag{36}$$

Substituting $\cos\chi = J/\sqrt{J^2+\cos^2\tau^*}$, $\sin\chi = -\cos\tau^*/\sqrt{J^2+\cos^2\tau^*}$, and the expression for $(v^*)^2$ (Equation 9), we get

$$F_n^* = \frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2\sqrt{J^2+\cos^2\tau^*}\;(|\cos\tau^*|\,C_L - J\,C_D) \tag{37}$$

$$F_s^* = -\frac{1}{2}\frac{A^*}{m^*}\left(s_0^*\omega^*\right)^2\sqrt{J^2+\cos^2\tau^*}\;(\sigma J\,C_L + \cos\tau^*\,C_D) \tag{38}$$

Writing $\theta \equiv \tan^{-1}(|\cos\tau^*|/J)$, such that $\alpha = \sigma\psi + \theta$ (see Equation 13),

$$\cos 2\theta = \frac{J^2 - \cos^2\tau^*}{J^2 + \cos^2\tau^*} \qquad \sin 2\theta = \frac{2J|\cos\tau^*|}{J^2 + \cos^2\tau^*} \tag{39}$$

and the aerodynamic coefficients (Equation 4 and Equation 5) become

$$C_L = C_{L,0}\left(\sigma\cos 2\theta\ \sin 2\psi + \sin 2\theta\ \cos 2\psi\right) \tag{40}$$

$$C_D = \bar{C}_D - \frac{\Delta C_D}{2}\left(\cos 2\theta\ \cos 2\psi - \sigma\sin 2\theta\ \sin 2\psi\right) \tag{41}$$

where we define $\bar{C}_D \equiv \left(C_{D,\frac{\pi}{2}} + C_{D,0}\right)/2$ for brevity. Substituting into the force components, using $\sigma|\cos\tau^*| = \cos\tau^*$ and $\sigma^2 = 1$, and averaging, the force coefficients are

$$\begin{aligned} C_{F_n^*} = {} & 2\left\langle \frac{\left[C_{L,0}\left(J^2 - \cos^2\tau^*\right) - \Delta C_D\, J^2\right]\cos\tau^*\ \sin 2\psi}{\sqrt{J^2+\cos^2\tau^*}} \right\rangle \\ & + 2\left\langle \frac{J\left[2C_{L,0}\cos^2\tau^* + \frac{\Delta C_D}{2}\left(J^2 - \cos^2\tau^*\right)\right]\cos 2\psi}{\sqrt{J^2+\cos^2\tau^*}} \right\rangle - 2J\bar{C}_D\left\langle\sqrt{J^2+\cos^2\tau^*}\right\rangle \end{aligned} \tag{42}$$

$$\begin{aligned} C_{F_s^*} = {} & -2\left\langle \frac{J\left[C_{L,0}\left(J^2 - \cos^2\tau^*\right) + \Delta C_D\cos^2\tau^*\right]\sin 2\psi}{\sqrt{J^2+\cos^2\tau^*}} \right\rangle \\ & - 2\left\langle \frac{\left[2C_{L,0}J^2 - \frac{\Delta C_D}{2}\left(J^2 - \cos^2\tau^*\right)\right]\cos\tau^*\ \cos 2\psi}{\sqrt{J^2+\cos^2\tau^*}} \right\rangle \end{aligned} \tag{43}$$

Note the constant ($\bar{C}_D$) drag term of $C_{F_s^*}$ vanishes because $\langle\cos\tau^*\sqrt{J^2+\cos^2\tau^*}\rangle = 0$. Setting $J = 0$ recovers Equation 20 and Equation 21, as expected. At the operating point $\delta_0 = 90°$, $\psi = \psi_0 - \psi_1\cos\tau^*$, and the parity argument of the hover case applies unchanged: under $\tau^* \to \tau^* + \pi$, the bracketed polynomials and $\sqrt{J^2+\cos^2\tau^*}$ are invariant, so expanding $\sin 2\psi$ and $\cos 2\psi$ and discarding the vanishing averages leaves

$$C_{F_n^*} = 2\, I_n(J,\psi_1)\ \cos 2\psi_0 - 2J\bar{C}_D\, I_0(J) \tag{44}$$

$$C_{F_s^*} = -2\, I_s(J,\psi_1)\ \sin 2\psi_0 \tag{45}$$

with

$$I_0 = \left\langle\sqrt{J^2+\cos^2\tau^*}\right\rangle \tag{46}$$

$$\begin{aligned} I_n = {} & \left\langle \frac{J\left[2C_{L,0}\cos^2\tau^* + \frac{\Delta C_D}{2}\left(J^2 - \cos^2\tau^*\right)\right]\cos\left(2\psi_1\cos\tau^*\right)}{\sqrt{J^2+\cos^2\tau^*}} \right\rangle \\ & - \left\langle \frac{\left[C_{L,0}\left(J^2 - \cos^2\tau^*\right) - \Delta C_D\, J^2\right]\cos\tau^*\ \sin\left(2\psi_1\cos\tau^*\right)}{\sqrt{J^2+\cos^2\tau^*}} \right\rangle \end{aligned} \tag{47}$$

$$\begin{aligned} I_s = {} & \left\langle \frac{J\left[C_{L,0}\left(J^2 - \cos^2\tau^*\right) + \Delta C_D\cos^2\tau^*\right]\cos\left(2\psi_1\cos\tau^*\right)}{\sqrt{J^2+\cos^2\tau^*}} \right\rangle \\ & + \left\langle \frac{\left[2C_{L,0}J^2 - \frac{\Delta C_D}{2}\left(J^2 - \cos^2\tau^*\right)\right]\cos\tau^*\ \sin\left(2\psi_1\cos\tau^*\right)}{\sqrt{J^2+\cos^2\tau^*}} \right\rangle \end{aligned} \tag{48}$$

The angle from the stroke-plane normal to the mean force (Equation 24) becomes

$$\beta = \tan^{-1}\left[\frac{I_s \sin 2\psi_0}{I_n \cos 2\psi_0 - J\bar{C}_D\, I_0}\right] \tag{49}$$

In the limit $J \to 0$, the I_0 term vanishes and

$$I_n \to C_{L,0}\,\langle \sin(2\psi_1 \cos\tau^*) \cos\tau^* |\cos\tau^*|\rangle \qquad I_s \to \frac{\Delta C_D}{2}\,\langle \sin(2\psi_1 \cos\tau^*) \cos\tau^* |\cos\tau^*|\rangle \tag{50}$$

recovering Equation 30. The overall force coefficient C_{F^*} is shown in Figure 5 for values of J from 0 to 1, at $\delta_0 = 90°$. Highlighted on each panel is the reference point from the hover plot Figure 3, tracked as it turns from a global maximum at $J = 0$ to a local maximum, and eventually to a saddle point. This point represents the maximum force available in the $+\hat{n}$ direction for a given J. The region of space within the dotted contour around the reference point corresponds to $C_{F_n^*} > 0$. Within this region, the stroke-plane-normal component of the mean force points towards $+\hat{n}$.

The values of C_{F^*} and ψ_1 at the reference point are shown as functions of J in Figure 6. As the figure makes clear, to obtain the maximum force along $+\hat{n}$, ψ_1 should be decreased as J increases. The mechanism behind this is illustrated in Figure 7. For a given ψ, α decreases as J is increased. This moves us away from the ideal α, reducing lift, and in turn the force component along $+\hat{n}$. By decreasing ψ, the ideal α is recovered, thrust along $+\hat{n}$ is maximized. Of course, this is an instantaneous picture, but the conclusion can be extended to the mean force over the wingbeat stroke. To achieve the required reduction in ψ everywhere throughout the stroke, it is hopefully clear that the relevant parameter is ψ_1.

Finally, Figure 8 shows β as a function of ψ_0 for J from 0 to 1. Specifically, since β now depends on ψ_1 in addition to ψ_0 (see Equation 49), each line is obtained using the ψ_1 value adapted to the value of J (as shown in Figure 5). As J is increased, there is a clear steepening of the slope around $\psi_0 = 0$, meaning control authority is significantly increased. This would tend to suggest that any gain associated with the control variable ψ_0 may need to adapt based on the current J.

We conclude this analytical section by summarizing the kinematic parameters for the general case, in Table 2. Since there is no requirement for steady flight like in hover, no specific values for the control variables s_0^* and ψ_0 are prescribed, however the relevant variables that would inform the choice of the values are still listed.

Table 2: Wing kinematic parameters (general case)

Parameter	Value	Variables
s_0^*	—	$\frac{A^*}{m^*}$ ω^* γ $C_{L,0}$ ΔC_D $\bar{C}_D$ J
ψ_0	—	γ $C_{L,0}$ ΔC_D $\bar{C}_D$ J
ψ_1	$\psi_1^{\text{opt}}(J)$	J $C_{L,0}$ ΔC_D
δ_0	90°	None

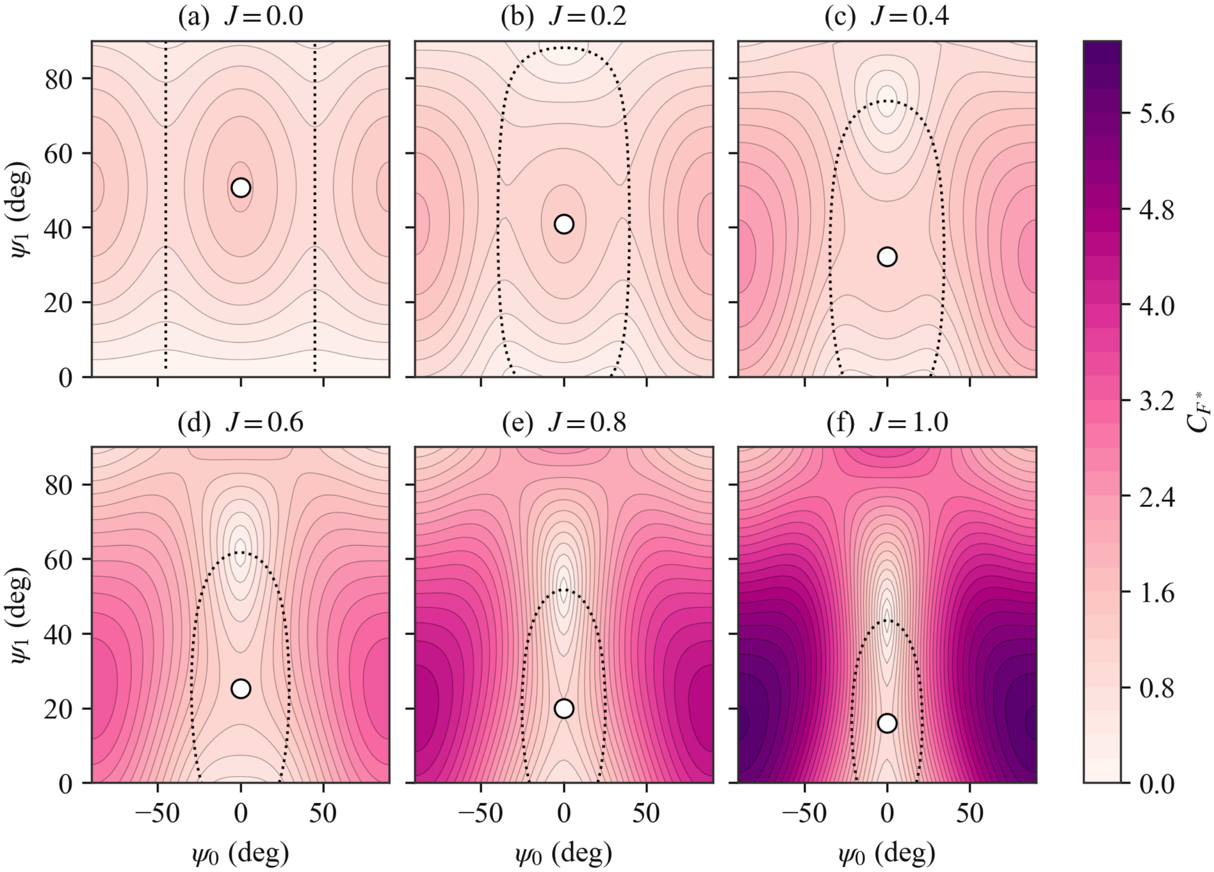

Figure 5: Force coefficient C_{F^*} over the pitch plane (ψ_0, ψ_1) at $\delta_0 = 90°$, for increasing advance ratio J. White dots track the hover optimum as it moves with J. It remains on the $\psi_0 = 0°$ axis by symmetry while ψ_1 decreases, and it turns from a local maximum into a saddle point. The global maximum sits at large $|\psi_0|$, where the force is dominated by drag from the normal inflow and points along $-\hat{n}$. The dotted contour marks $C_{F_n^*} = 0$, the boundary between propulsive ($+\hat{n}$) and braking ($-\hat{n}$) mean force.

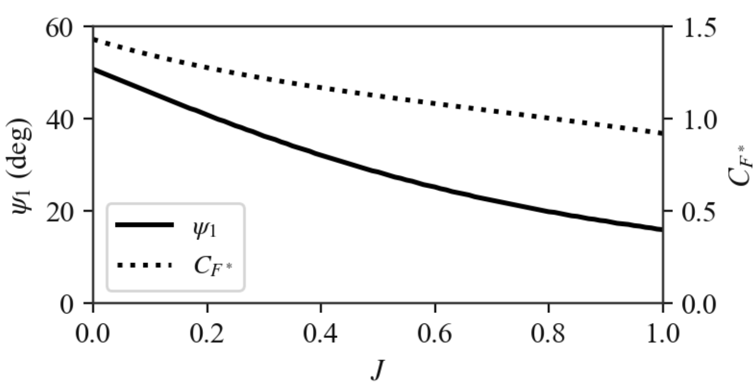

Figure 6: Pitch amplitude ψ_1 and force coefficient C_{F^*} of the continued hover optimum (the point tracked in Figure 5) as functions of advance ratio J. The point turns from a local maximum into a saddle at $J \approx 0.46$.

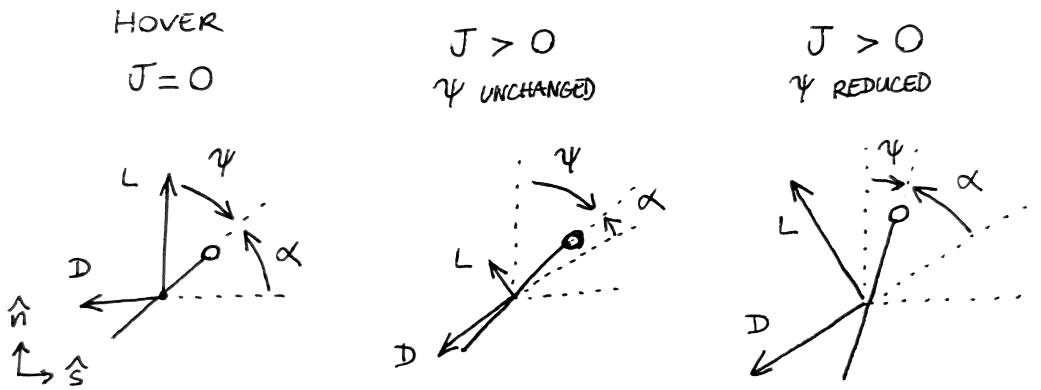

Figure 7: Illustration of the ψ reduction mechanism for maximizing stroke-plane-normal force as J increases.

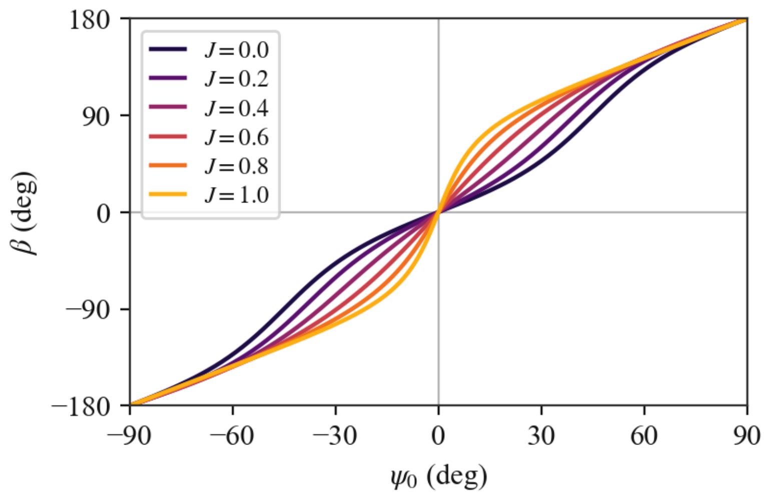

Figure 8: Effect of ψ_0 on the force direction angle β for increasing advance ratio J, at $\delta_0 = 90°$ and with ψ_1 set, for each J, to the continued-optimum value of Figure 6. Unlike in the hover case, β depends on ψ_1 when $J > 0$.

3.3 Control Strategy

From the above analysis, we propose a control strategy. It will be discussed qualitatively in this section, and formalized in the following section (subsection 3.4). We begin by going over *what to do* with each of the wing kinematic parameters.

- *Stroke plane angle γ:* when hovering, $\gamma = \gamma^{\text{hover}}$. When moving, γ should change to effectively obtain, or at least approximate, the axial inflow we have assumed in the analysis. There is the issue of the transition between these two phases, which may be abrupt if the commanded flight direction is not aligned with the stroke-plane normal at hover. We propose that γ be varied smoothly from its hover value to its moving flight value based on J.

- *Stroke amplitude s_0^*:* primary control for the mean force magnitude, should be varied so that the force magnitude demand is met. In hovering or in general steady flight, this means having the mean force magnitude equal the weight.

- *Mean pitch ψ_0:* primary control for the mean force direction relative to the stroke-plane normal, should be varied so that the force direction demand is met. In hovering or in general steady flight, this means having the mean force vector point toward $+\hat{z}$.
- *Pitch amplitude ψ_1:* should be varied with J to maximize the available force, as shown in Figure 6.
- *Pitch phase δ_0:* fixed to 90°, not a control variable.
- *Wingbeat frequency ω^*:* constant parameter, not a control variable.

In summary, the two variables that directly act to modulate and direct the force are s_0^* and ψ_0. The two remaining variables, γ and ψ_1, adapt to the velocity vector to respectively approximate an axial inflow, and maximize the available force.

A comment should be made here about the $\hat{x}$-wise asymmetry introduced by the nonzero γ^{hover}. At low speeds, when the stroke plane is still close to its hover orientation, this will result in different behaviors for motion toward $+\hat{x}$ and $-\hat{x}$. To restore symmetry, we propose to flip the sign of γ^{hover} when the commanded flight direction changes.

3.4 Control Law

We now formalize the strategy of subsection 3.3. Let $\vec{x}^*$ and $\vec{u}^*$ be the body position and velocity. In general, a prescribed trajectory is characterized at every point by a reference position $\vec{x}_r^*$, velocity $\vec{u}_r^*$, and acceleration $\vec{a}_r^*$ (*e.g.* centripetal acceleration for a circular trajectory). Depending on the exact requirement for the controller, only some of these inputs may be necessary for the control law. In the case of hovering for example, if the objective is to hold a specific position in space, $\vec{x}_r^*$ will be needed, but $\vec{u}_r^*$ is only needed to provide additional damping through derivative control. If the objective is simply to hover about an arbitrary point, then only $\vec{u}_r^*$ is needed. Similarly, in the case of prey pursuit, the input is the line-of-sight bearing to the prey, without range information, and the objective is to align the velocity vector with the line of sight as we close in on the prey. In this case, only $\vec{u}_r^*$ needs to be formulated, which prescribes both the pursuit direction (line of sight) and the desired closing speed. With this in mind, we start by forming the force demand in the most general case, using all the potential inputs. The result is a proportional-derivative tracking law with acceleration feedforward and compensation of the weight

$$\vec{a}_{\text{des}}^* = \vec{a}_r^* + K_p\left(\vec{x}_r^* - \vec{x}^*\right) + K_d\left(\vec{u}_r^* - \vec{u}^*\right) \tag{51}$$

$$\langle \vec{F}^* \rangle^{\text{des}} = \vec{a}_{\text{des}}^* + \hat{z} \tag{52}$$

In the hover and prey pursuit cases, $\vec{a}_r^* = \vec{0}$. In the prey pursuit case specifically, $K_p = 0$, while either K_p or K_d may be zero in the hover case. We will see that in practice, a simpler velocity-based expression ($\vec{a}_{\text{des}}^* = K_d\left(\vec{u}_r^* - \vec{u}^*\right)$) is very satisfactory. With the force demand formulated, γ and ψ_1 must now adapt to the measured velocity. The advance ratio is evaluated as

$$J = \frac{|\vec{u}^*|}{\left(s_0^*\omega^*\right)^{\text{ref}}} \tag{53}$$

where $\left(s_0^*\omega^*\right)^{\text{ref}}$ is the wingbeat velocity of the reference configuration at the hover condition, held fixed so that J does not depend on the control variable s_0^*. The stroke-plane angle blends from the inclined-hover value to the velocity-aligned value γ_u, defined such that $\hat{n}$ is aligned with $\vec{u}^*$:

$$\gamma = [1 - f(J)]\,\sigma_x\gamma^{\text{hover}} + f(J)\,\gamma_u \qquad f(J) = 1 - e^{-(J/J_c)^2} \tag{54}$$

with $J_c = 0.35$, and where $\sigma_x \in \{-1, +1\}$ is the sign of the commanded horizontal heading, latched at its most recent nonzero value so that the stroke plane does not flip on the residual hover velocity (see subsection 3.3). The pitch amplitude is slaved to the force-maximizing value of Figure 6,

$$\psi_1 = \psi_1^{\text{opt}}(J) \tag{55}$$

under an assumed axial inflow. The actual inflow deviates from the stroke-plane normal during transients, but only appreciably at low advance ratio, where the optimum is insensitive to the inflow direction, so the deviation is neglected. With (γ, ψ_1) now set, the remaining two variables (s_0^*, ψ_0) are trimmed to the force demand by solving

$$\langle \vec{F}^* \rangle \left(s_0^*, \psi_0;\, \gamma, \psi_1, J\right) = \langle \vec{F}^* \rangle^{\text{des}} \tag{56}$$

for (s_0^*, ψ_0), using the expressions for $\langle \vec{F}^* \rangle$ derived in subsection 3.2. Finally, once the reference trajectory is exhausted and the position error falls within a small radius (0.3 body lengths), γ and ψ_1 are pinned to their hover values $\sigma_x \gamma^{\text{hover}}$ and $\psi_1^{\text{opt}}(0)$, avoiding jitter from the residual hover velocity, while Equation 56 continues to hold hover equilibrium.

4 Results

4.1 Reference Configuration

To ground our results, we introduce a reference configuration based on morphological and wingbeat frequency data for *Anisoptera* from [4] and [5], respectively. Nondimensionalized, the data range is [0.2, 1.0] for A^*/m^*, and [8, 20] for ω^*. For the reference configuration, we select the intermediate values

$$\frac{A^*}{m^*} = 0.6 \qquad\qquad \omega^* = 14 \tag{57}$$

In addition, we choose the hovering stroke plane angle to be

$$\gamma^{\text{hover}} = -45^\circ \tag{58}$$

defined with a negative sign (clockwise) so that it corresponds to a horizontal dragonfly pointing towards $+\hat{x}$. When settling from lateral motion towards $-\hat{x}$, the hovering stroke plane angle would be $-\gamma^{\text{hover}} = +45^\circ$, as explained in subsection 3.3.

Since dragonflies have two wing pairs, we set $N = 2$, and the wing pair phase shift parameter is set to $\sigma_0 = 180^\circ$ so as to reduce body oscillations.

4.2 Hover

The hover regime of the control law is exercised by initializing the body at the reference position, $\vec{x}^*(0) = \vec{x}_r^* = (0, 0)$, with a velocity perturbation $\vec{u}^*(0) = (0.2, 0.2)$ and $\vec{u}_r^* = 0$. Position feedback only ($K_d = 0$), velocity feedback only ($K_p = 0$), and the full tracking law are compared in Figure 9, Figure 10, and Figure 11, with $K_p = 2.25$ and $K_d = 2.7$.

4.3 Prescribed Trajectory

We choose an upward bend as the reference trajectory: a horizontal run of 1.5 body lengths, a circular arc of radius 2 bending up by 45°, and a straight climb of 1.5 body lengths, traversed with a trapezoidal speed schedule (linear ramps from and to rest over a distance of 0.3, cruise speed 0.7). The reference acceleration $\vec{a}_r^*$

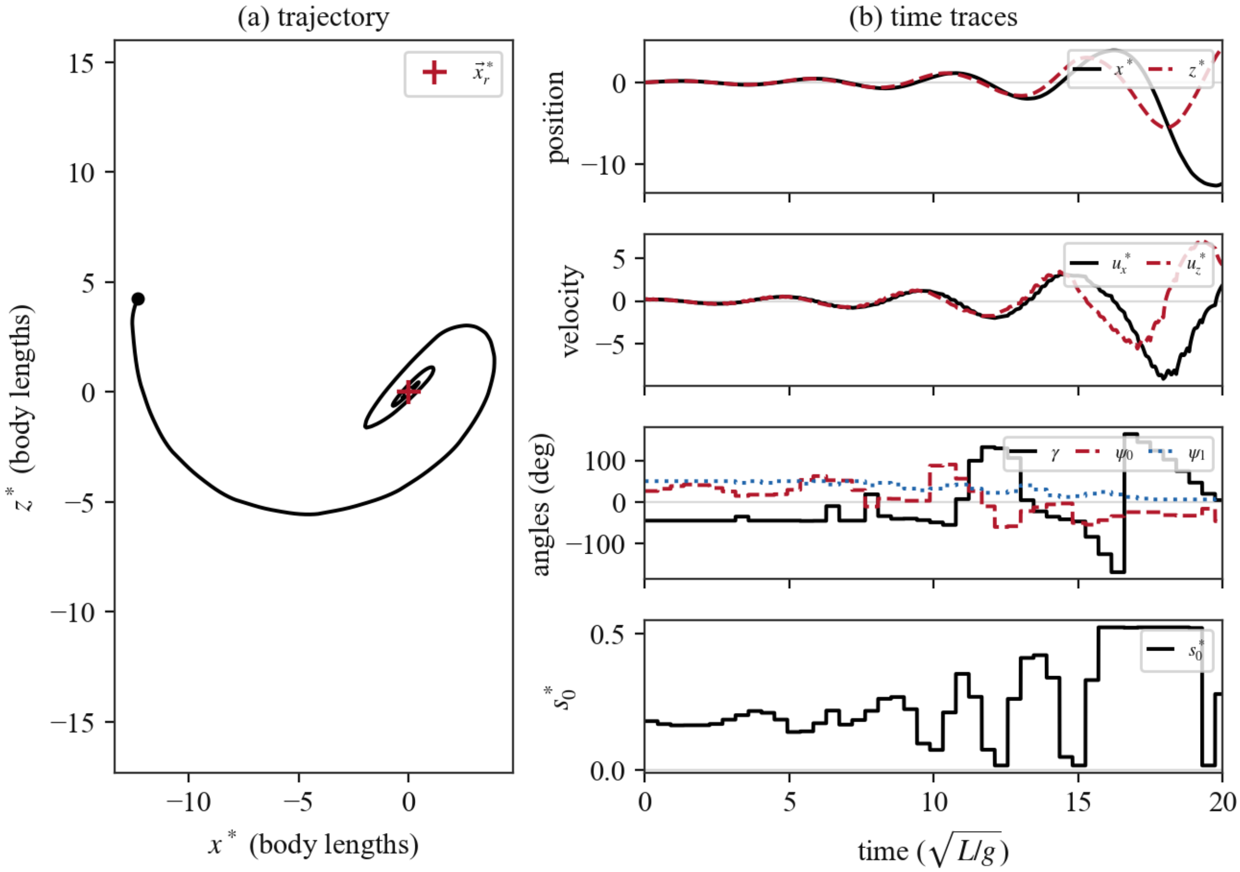

Figure 9: Hover hold with position feedback only ($K_d = 0$). The oscillation about $\vec{x}_r^*$ is not damped, and the phase lag of the once-per-wingbeat control update makes it grow. Once the position error exceeds the hover pin radius, γ and ψ_1 resume scheduling and the trim saturates.

carries the tangential (ramp) and centripetal (arc) demand. Starting from rest on the path, four configurations are compared: position feedback only (Figure 12), velocity feedback only (Figure 13), both (Figure 14), and both with acceleration feedforward—the full law of Equation 52 (Figure 15)—again with $K_p = 2.25$ and $K_d = 2.7$.

4.4 Prey Pursuit

Prey pursuit closes the guidance loop through the control law using only the velocity term of Equation 52. The reference velocity is defined as the desired closing speed along the line of sight to the prey,

$$\vec{u}_r^* = u_{\text{cmd}}^* \, \hat{r} \qquad \hat{r} = \frac{\vec{x}_{\text{prey}}^* - \vec{x}^*}{|\vec{x}_{\text{prey}}^* - \vec{x}^*|} \tag{59}$$

with no position reference or acceleration feedforward, so that $\langle \vec{F}^* \rangle^{\text{des}} = K_d(\vec{u}_r^* - \vec{u}^*) + \hat{z}$. The prey starts at $(2.5, 2.0)$ and moves at the constant velocity $(0.35, 0)$; the body starts at rest at the origin, with $u_{\text{cmd}}^* = 0.7$ and $K_d = 2.7$. Interception is declared at a range of 0.5 body lengths. The resulting pursuit is shown in Figure 16.

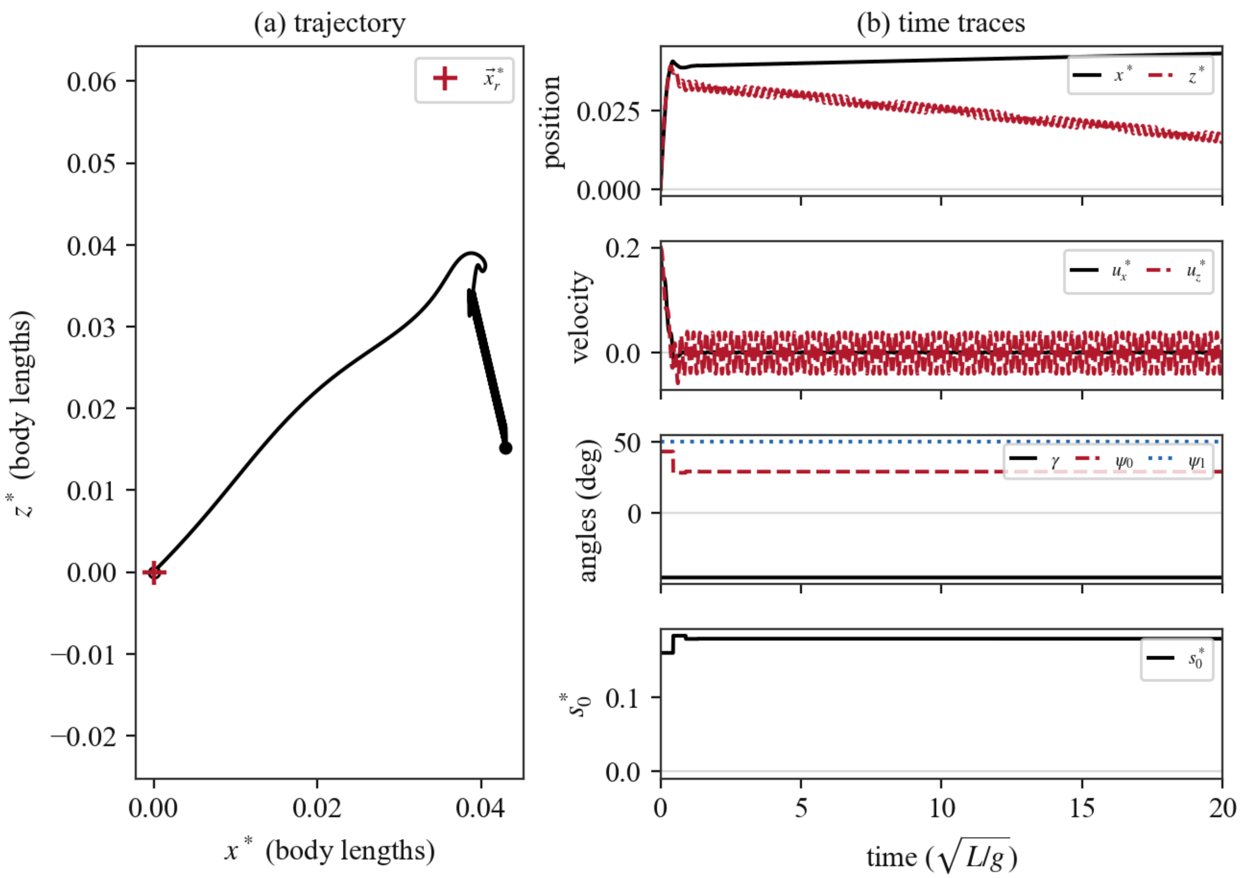

Figure 10: Hover hold with velocity feedback only ($K_p = 0$). The velocity perturbation is eliminated within about one time unit, but the position offset accumulated meanwhile (about 0.05 body lengths) is never corrected. With that said, the resulting error is very small, and while there is some residual drift over time, the result is quite satisfactory.

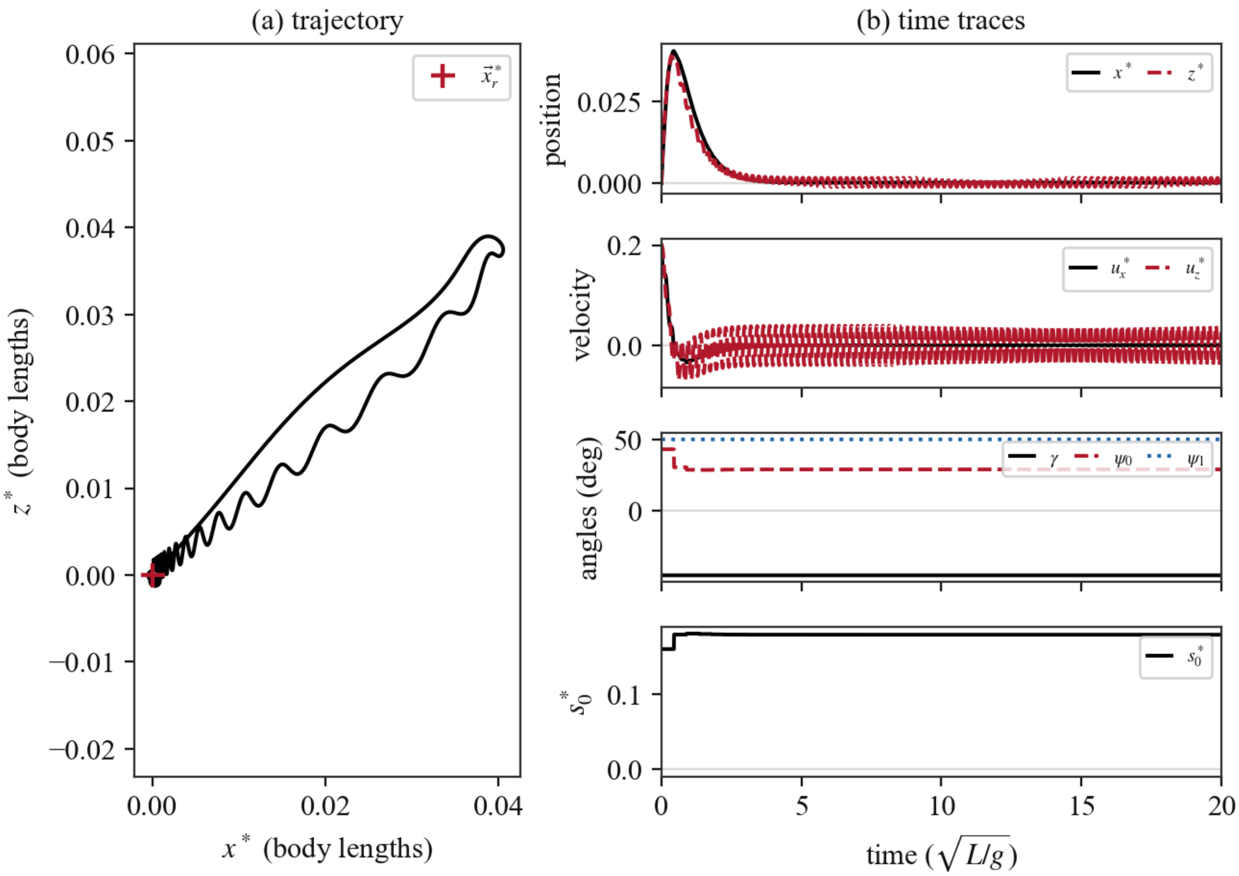

Figure 11: Hover hold with the full tracking law. The body returns to $\vec{x}_r^*$ (peak excursion 0.06 body lengths, residual position error below 2×10^{-3}).

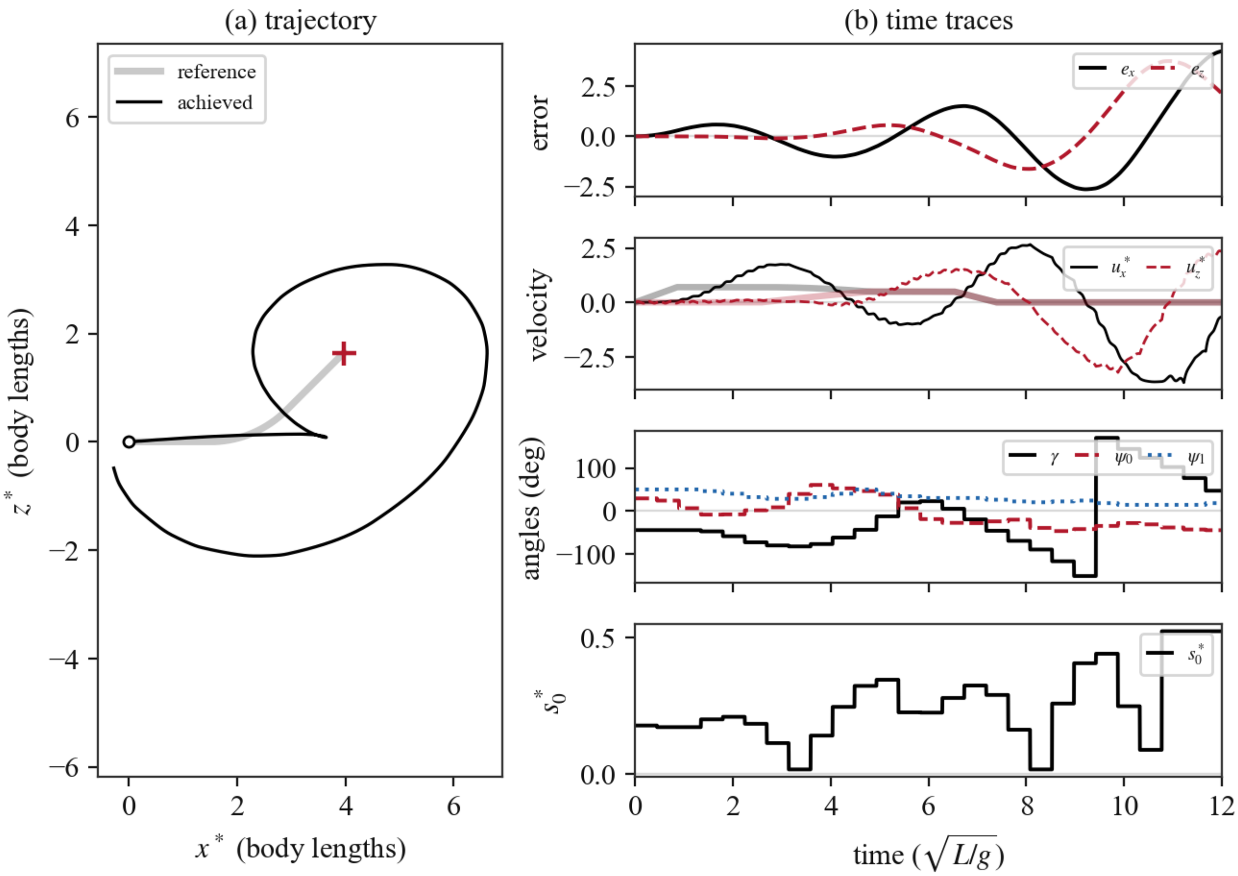

Figure 12: Prescribed trajectory with position feedback only ($K_d = 0$): as in hover, the undamped error dynamics under the once-per-wingbeat update are unstable, and the body departs from the path.

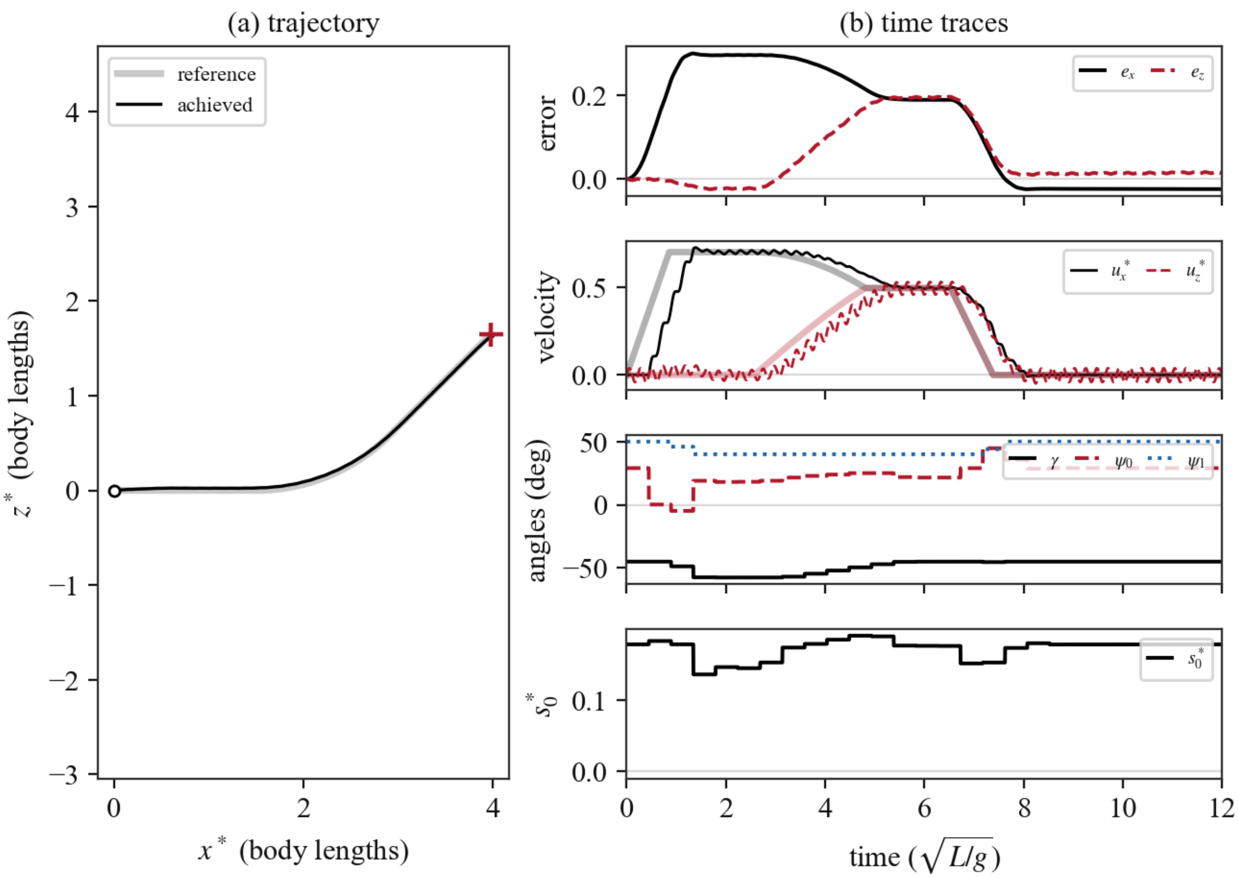

Figure 13: Prescribed trajectory with velocity feedback only ($K_p = 0$): the body follows the commanded velocity with a first-order lag, trailing the reference by $|\vec{u}_r^*|/K_d \approx 0.26$ body lengths throughout. The deceleration ramp lets it catch up near the end, but nothing corrects position drift. With that said, the resulting trajectory appears very satisfactory.

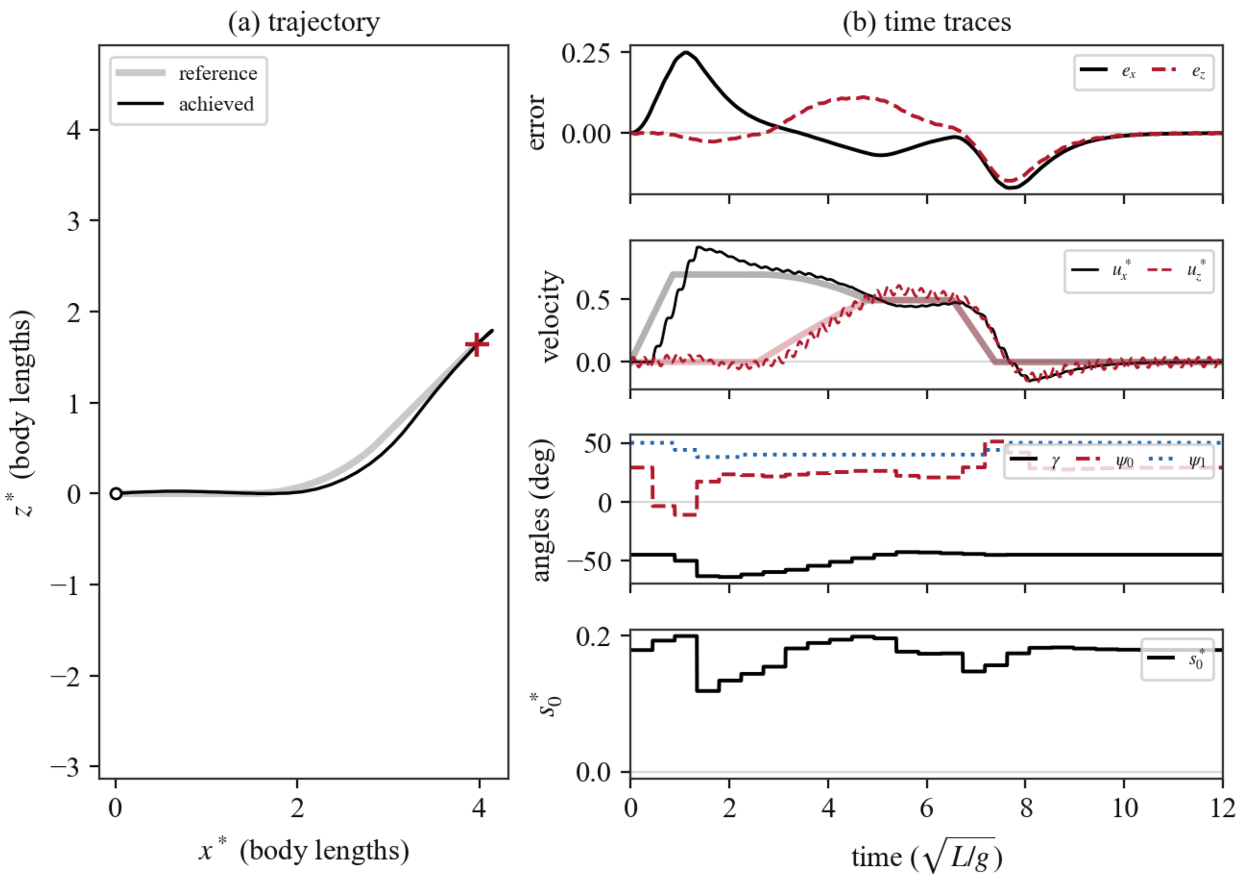

Figure 14: Prescribed trajectory with position and velocity feedback ($K_p + K_d$, no feedforward). Tracking error peaks at 0.25 body lengths, dominated by the speed-ramp lag and by the "cutting" inside the arc, since the required acceleration must be generated by a standing error.

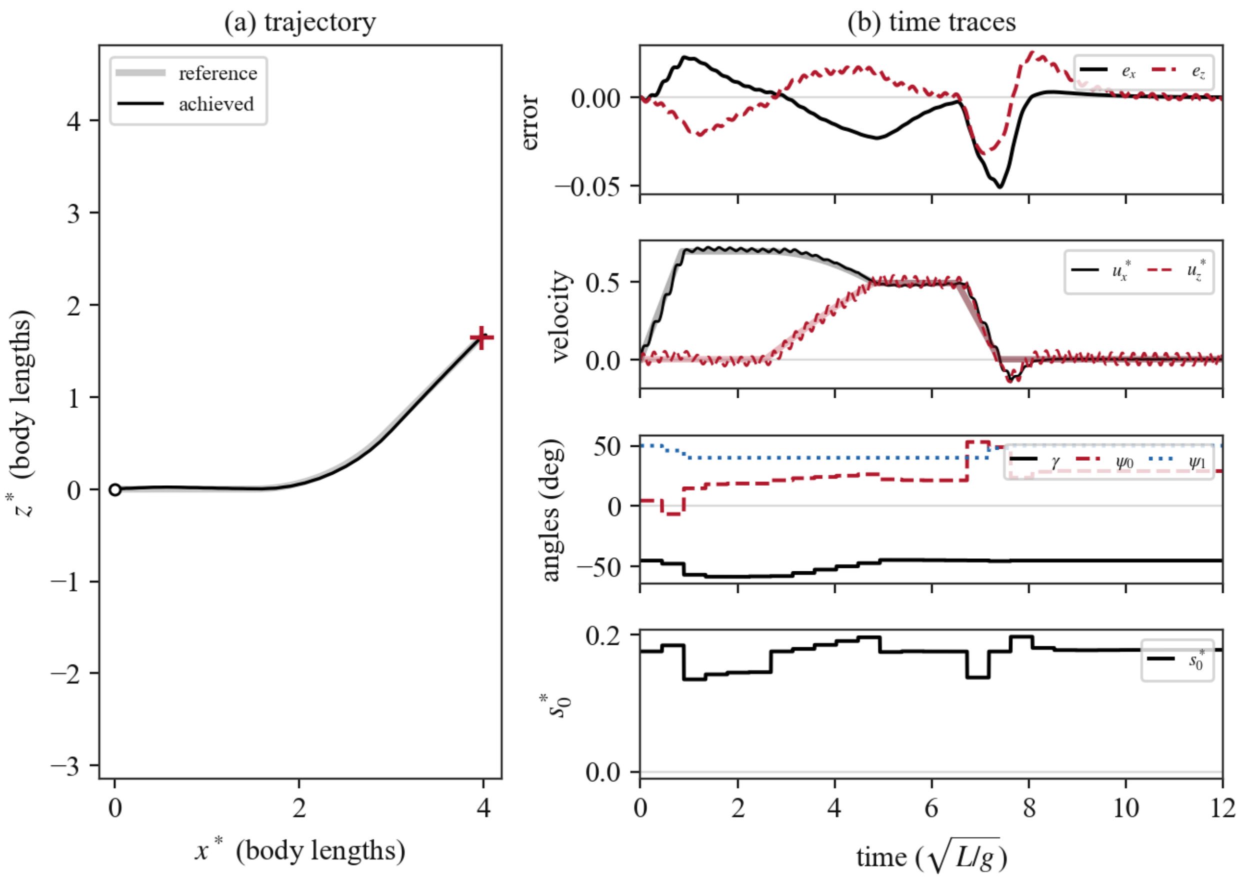

Figure 15: Prescribed trajectory with the full law of Equation 52, including the acceleration feedforward $\vec{a}_r^*$. The maneuver demand is supplied directly rather than through the error terms, and the peak tracking error drops significantly.

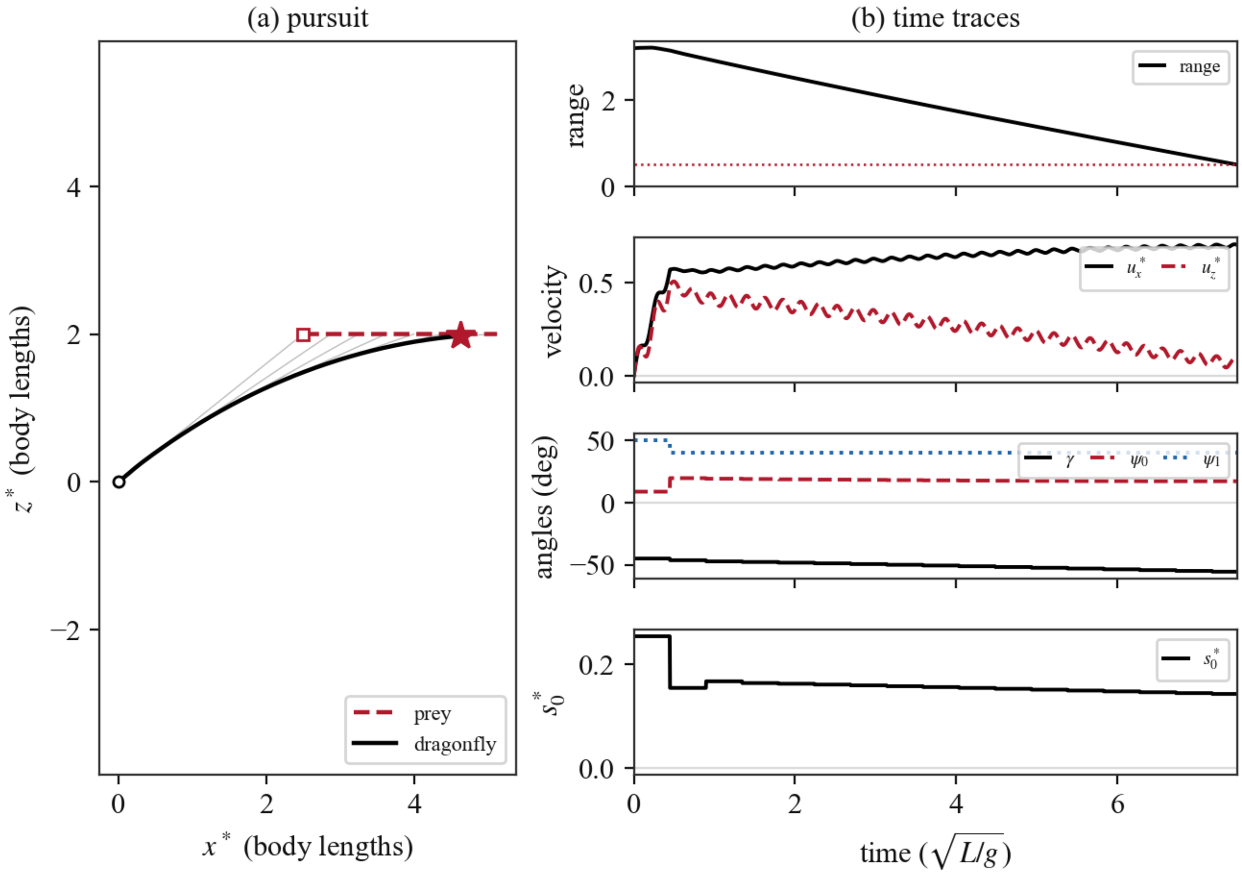

Figure 16: Prey pursuit. (a) The line-of-sight command produces the classical pure-pursuit tail chase, with capture at $t^* \approx 7.5$ (star). (b) The body accelerates to the commanded closing speed ($J \approx 0.2$). γ leans from γ^{hover} toward alignment with the flight direction, ψ_1 follows its slaved schedule (Equation 55), and (s_0^*, ψ_0) trim the force demand.

5 Conclusion

The proposed control scheme is capable of achieving hover and prey pursuit, and of following arbitrary trajectories to a reasonable degree of accuracy. Some caveats are worth mentioning:

- *Not strictly proportional control*. While all control tasks can be achieved with a single proportional gain K_d in the force demand formulation, the control variables vary nonlinearly with the force demand, since we are solving the inverse problem of which variable values satisfy the force demand. Following the hover controller presented in the previous report, we may be able to resolve this by linearizing the force, using the expressions we derived in subsection 3.2.

- *Issue with descent*. When descending (towards $-\hat{z}$) at a fast rate, the stroke-plane normal will point downwards. However, to achieve a steady descent velocity, the force vector must point up, opposed to gravity. The force vector, then, must point toward $-\hat{n}$. This can be achieved as we saw in subsection 3.2 with large enough $|\psi_0|$, but at high J it results in large force magnitudes which lead to instability in the unsteady dynamics, and prevent clean descent trajectories. This could potentially be addressed with an alternate control mode using gliding instead of flapping to achieve steady descent. Note that descent is not as problematic if the commanded trajectory has a sustained downward acceleration larger than that due to gravity, since the requirement that the force oppose gravity does not hold.

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